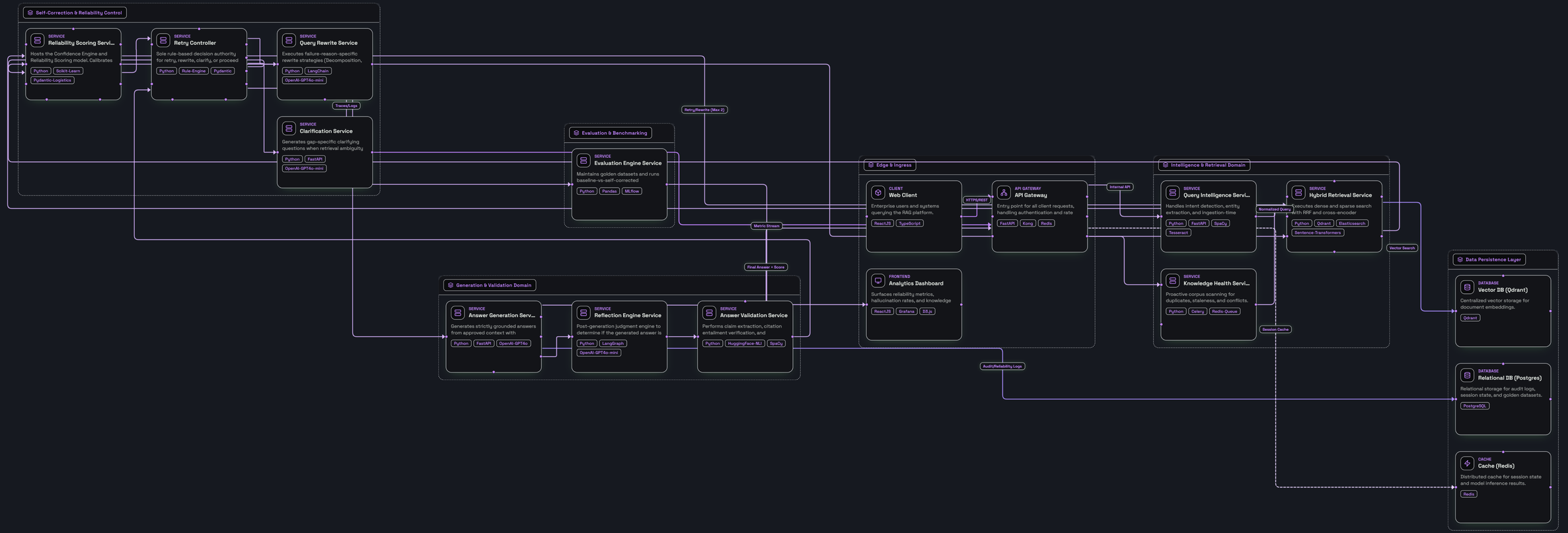
Self-Correction & Reliability Control
SERVICE
Reliability Scoring Servi...
Hosts the Confidence Engine and Reliability Scoring model. Calibrates
Python
Scikit-Learn
Pydantic-Logistics
SERVICE
Retry Controller
Sole rule-based decision authority for retry, rewrite, clarify, or proceed
Python
Rule-Engine
Pydantic
SERVICE
Query Rewrite Service
Executes failure-reason-specific rewrite strategies (Decomposition,
Python
LangChain
OpenAI-GPT4o-mini
Traces/Logs
SERVICE
Clarification Service
Generates gap-specific clarifying questions when retrieval ambiguity
Python
FastAPI
OpenAI-GPT4o-mini
Retry/Rewrite (Max 2)
Evaluation & Benchmarking
SERVICE
Evaluation Engine Service
Maintains golden datasets and runs baseline-vs-self-corrected
Python
Pandas
MLflow
Metric Stream
Edge & Ingress
CLIENT
Web Client
Enterprise users and systems querying the RAG platform.
React.JS
TypeScript
HTTPS/REST
API GATEWAY
API Gateway
Entry point for all client requests, handling authentication and rate
FastAPI
Kong
Redis
Internal API
FRONTEND
Analytics Dashboard
Surfaces reliability metrics, hallucination rates, and knowledge
React.JS
Grafana
D3.js
Intelligence & Retrieval Domain
SERVICE
Query Intelligence Servi...
Handles intent detection, entity extraction, and ingestion-time
Python
FastAPI
SpaCy
Tesseract
Normalized Query
SERVICE
Hybrid Retrieval Service
Executes dense and sparse search with RRF and cross-encoder
Python
Qdrant
Elasticsearch
Sentence-Transformers
Vector Search
SERVICE
Knowledge Health Servi...
Proactive corpus scanning for duplicates, staleness, and conflicts.
Python
Celery
Redis-Queue
Session Cache
Final Answer + Score
Generation & Validation Domain
SERVICE
Answer Generation Serv...
Generates strictly grounded answers from approved context with
Python
FastAPI
OpenAI-GPT4o
SERVICE
Reflection Engine Service
Post-generation judgment engine to determine if the generated answer is
Python
LangGraph
OpenAI-GPT4o-mini
SERVICE
Answer Validation Service
Performs claim extraction, citation entailment verification, and
Python
HuggingFace-NLI
SpaCy
Audit/Reliability Logs
Data Persistence Layer
DATABASE
Vector DB (Qdrant)
Centralized vector storage for document embeddings.
Qdrant
DATABASE
Relational DB (Postgres)
Relational storage for audit logs, session state, and golden datasets.
PostgreSQL
CACHE
Cache (Redis)
Distributed cache for session state and model inference results.
Redis